Homework 29.10.2026

↳ prop. 2.4.1 from theory; exercise 6.2 c

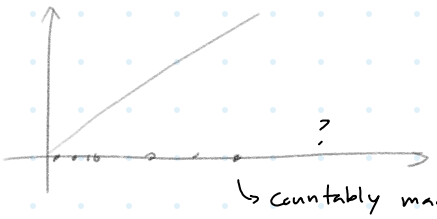
Proposition 2.4.1. Let $\mathcal{C}$ be the set of all intervals of the form $(-\infty, a]$ in $\mathbb{R}$. Then $\mathcal{B}(\mathcal{C}) = \mathcal{B}_{\mathbb{R}}$

↳ proof:

The Borel algebra $\mathcal{B}_{\mathbb{R}}$ is defined as $\mathcal{B}(\mathcal{D})$ where $\mathcal{D}$ is the set of all open intervals in $\mathbb{R}$ i.e. $\mathcal{D} = \{(a,b);\ a,b \in \mathbb{R}\}$,
while $\mathcal{C} = \{(-\infty, a];\ a \in \mathbb{R}\}$. We need to show these two sets generate the same $\mathcal{B}$-alg.
Any $\mathcal{B}$-alg. $\mathcal{B}$ must for $A_i \in \mathcal{B}$ also contain A_1^c and $\overset{\infty}{\underset{i}{\cup}} A_i$
since $(a,b)^c = (-\infty, a] \cup [b, \infty)$, $(-\infty, a] \cup [b, \infty) \in \mathcal{B}_{\mathbb{R}}$

$(-\infty, a]^c = (a, \infty) = \overset{\infty}{\underset{n=1}{\cup}} (a, n)$ ... IDK

6.2 c) Find the Lebesgue integral of $f(\omega) = \begin{cases} n; & \omega = \mathbb{Q}^c \cap [0, n] \\ 0; & \text{elsewhere} \end{cases}$ on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \lambda)$



↳ countably many points with $f(\omega) = 0$

$$\int f(\omega)\, d\lambda = n \cdot \lambda(\mathbb{Q}^c \cap [0, n]) = n \cdot n = \underline{\underline{n^2}}$$